

INDU 211: TERM PAPER GROUP PRESENTATION GUIDE & 15-MINUTE VIDEO SCRIPT

"The Future of Industrial Engineering in the Generative AI Era"

Department of Mechanical, Industrial and Aerospace Engineering · Concordia University

Course: INDU 211 / Section 2X — Introduction to Production and Manufacturing Systems

Due Date: Thursday, December 1, 2026 (on Moodle)

Format: 15-Minute Recorded Video Presentation (Zoom / Microsoft Teams / OBS Studio)

1. Video Recording Guidelines & Team Plan

Parameter	Official Specification
Target Duration	Exactly 15 minutes (between 14:30 and 15:30)
Team Participation	Every team member must present an equal share (6 members)
Platform	Zoom or Microsoft Teams (screen share + webcams active)
Format	MP4 video recording submitted on Moodle
Style	Clear, natural speaking; professional pacing; do not read slides verbatim

Team Speaker Schedule (6 Members · 2 Minutes 30 Seconds Each)

- **Speaker 1 (00:00 – 02:30):** Title & team introductions, what is Industrial Engineering, systems conversion model, and rise of GenAI.
- **Speaker 2 (02:30 – 05:00):** Traditional Predictive AI vs. Generative AI, and how daily factory and office workplaces are transforming.
- **Speaker 3 (05:00 – 07:30):** Automation potential in production operations: scheduling, facility layout planning, and assembly line balancing.
- **Speaker 4 (07:30 – 10:00):** Automation potential in inventory control, automated quality inspection (computer vision & SPC), and ergonomics.
- **Speaker 5 (10:00 – 12:30):** Critical AI risks: hallucinations, physical common-sense deficits, black-box opacity, and OIQ legal liability.
- **Speaker 6 (12:30 – 15:00):** Why human engineers are irreplaceable (Human-in-the-Loop), essential competencies for future graduates, and conclusion.

2. Slide-by-Slide Presentation Script

Slide 1: Title & Team Introductions (00:00 – 01:15)

- **On Screen:** Slide with project title, course name (INDU 211), Concordia University logo, and names of all 6 team members.
- **Speaker 1 (Team Leader):**

"Hello everyone, and welcome to our presentation. Today, our group will be presenting on 'The Future of Industrial Engineering in the Generative AI Era: Opportunities, Limitations, and the Role of Human Engineers'.

My name is [Name 1], and I am presenting today alongside my five colleagues: [Name 2], [Name 3], [Name 4], [Name 5], and [Name 6]. Together, we are Team 4 for INDU 211.

Over the next fifteen minutes, our team will explore how artificial intelligence is changing modern production systems. We will examine what tasks AI can automate, where AI systems fail, and why human industrial engineers remain essential to safe, productive manufacturing."

Slide 2: What is Industrial Engineering & Systems Thinking (01:15 – 02:30)

- **On Screen:** Conversion Process Model diagram: Inputs (People, Machines, Materials, Energy) → Transformation Process → Outputs (Products, Services). Overlay showing IE goals: Waste Reduction, Productivity, Quality, and Worker Safety.
- **Speaker 1 (Team Leader):**

"To build our foundation, let's revisit what industrial engineers actually do. Unlike other disciplines that focus solely on individual components, industrial engineers design and optimize complete integrated systems.

As we learned in INDU 211, every manufacturing operation follows the Conversion Process Model: transforming raw inputs—materials, labor, energy, and information—into finished goods and services. Our goal is to eliminate waste, improve throughput, maintain quality, and ensure worker safety across strategic, tactical, and operational control levels.

Recently, tools like ChatGPT, machine learning, and generative design have entered engineering offices. Will these tools replace industrial engineers, or empower them? I will now pass the presentation to [Name 2] to break down the core technology."

Slide 3: Traditional AI vs. Generative AI in Production (02:30 – 03:45)

- **On Screen:** Comparative table:
 - **Traditional AI:** Numerical pattern recognition, regression, predictive maintenance, demand forecasting.
 - **Generative AI:** Unstructured content generation, LLMs writing code/SOPs, diffusion/generative CAD design models.
- **Speaker 2:**

"Thank you, [Name 1]. Before looking at shop-floor impact, we must clearly distinguish between traditional predictive AI and modern Generative AI.

Traditional AI has been used in manufacturing for years. It processes structured numerical data to find patterns and make predictions. A classic example is Predictive Maintenance: sensors monitor motor vibration, and machine learning models predict bearing failure before a breakdown occurs. Another example is statistical demand forecasting.

Generative AI, on the other hand, is a breakthrough because it creates novel content from unstructured prompts. Instead of outputting a single number, LLMs can draft standard operating procedures, generate Python scripts for simulation, summarize maintenance logs, and suggest initial 3D layout geometries. Think of traditional AI as an ultra-fast analytical calculator, and Generative AI as a creative digital drafting assistant."

Slide 4: How AI Tools are Transforming Daily Workplaces (03:45 – 05:00)

- **On Screen:** Visual workflow diagram highlighting 3 major workplace shifts: (1) Automated shift reporting & documentation, (2) Accelerated natural-language data querying, (3) Real-time troubleshooting co-pilots on technician tablets.
- **Speaker 2:**

"So how is this changing day-to-day work in manufacturing facilities?

First, it drastically reduces routine paperwork. Industrial engineers frequently spend hours documenting shift handovers, root-cause incident reports, and standard operating procedures. Generative AI can synthesize raw technician notes into polished, standardized engineering documentation in seconds.

Second, it accelerates data analysis. Rather than constructing complicated SQL queries or spreadsheet macros, an engineer can query a database in natural language: 'Show me the scrap rate and bottleneck stations for Line 3 this week,' and receive immediate visualizations.

Third, shop-floor technicians now use interactive mobile co-pilots. When an error code appears on a CNC machine, AI cross-references equipment manuals and historical repair logs to guide troubleshooting.

Now, [Name 3] will examine which core operations can be automated."

Slide 5: Automation in Production Scheduling & Sequencing (05:00 – 06:15)

- **On Screen:** Gantt chart and sequencing matrix showing job-shop machine assignment, setup time minimization algorithms, and dispatching rules (SPT, EDD, Johnson's Rule).
- **Speaker 3:**

"Thanks, [Name 2]. When we look across the INDU 211 curriculum, different areas display vastly different potential for automation.

Production scheduling is one of the highest-potential areas. Determining optimal job sequencing to minimize makespan, balance machine utilization, and reduce setup changeover times is fundamentally a mathematical optimization challenge.

Algorithms, genetic heuristics, and AI solvers can evaluate millions of schedule permutations in seconds—tasks that would take human planners days. AI dynamically re-sequences job queues when priority rush orders arrive.

However, algorithmic schedules often fail when unexpected shop-floor events occur—such as an operator absence or an unrecorded tool fracture. While AI generates the optimized schedule, a human supervisor must validate its real-world viability."

Slide 6: Automation in Facility Layout Design & Line Balancing (06:15 – 07:30)

- **On Screen:** Factory floor layout schematic: Algorithmic block layout vs. actual physical constraints (support columns, ceiling heights, emergency exits, and material handling pathways).
- **Speaker 3:**

"Another major area is facility layout planning and assembly line balancing.

Generative layout algorithms can rapidly iterate through hundreds of block layout arrangements to minimize material handling transport costs, using principles similar to the CRAFT and ALDEP methods we study in class.

Similarly, in assembly line balancing, AI algorithms assign work elements to stations to minimize balance delay and cycle idle time.

Yet here is the crucial limitation: AI algorithms generate solutions in idealized 2D space. They do not know that a structural pillar blocks forklift access, that floor load capacity cannot support a heavy press, or that workers need safe access to emergency exits. Layouts proposed by AI always require physical ground-truthing by an engineer.

Next, [Name 4] will discuss inventory, quality, and ergonomics."

Slide 7: Automation in Inventory Control & Supply Chains (07:30 – 08:45)

- **On Screen:** Inventory sawtooth model with dynamic reorder points, Economic Order Quantity (EOQ), safety stock bands, and multi-echelon supply chain visibility.
- **Speaker 4:**

"Thank you, [Name 3]. Turning to inventory control, AI is transforming how factories manage raw materials and work-in-process stock.

Traditional inventory management relies on static formulas like the basic EOQ and fixed reorder points. In contrast, modern AI models dynamically adjust reorder thresholds by factoring in real-time supply chain lead times, seasonal demand fluctuations, and supplier reliability indices.

Automated warehouse systems use AI to track stock movements with RFID sensors, optimizing storage slotting and triggering purchase orders without manual human intervention.

Nonetheless, when macro-level supply disruptions occur—such as global shipping bottlenecks or supplier bankruptcies—AI historical models break down. Human engineers must step in to negotiate alternate sourcing, re-engineer product bills of materials, and adjust production plans."

Slide 8: Automated Quality Control (SPC / Vision) & Workplace Ergonomics (08:45 – 10:00)

- **On Screen:** High-speed camera inspection conveyor belt with defect bounding boxes, $\bar{X}$ and R control charts with automated out-of-control rule triggers, and ergonomic posture tracking.
- **Speaker 4:**

"In quality control, computer vision and automated Statistical Process Control (SPC) have reached very high automation levels.

High-speed vision systems inspect thousands of parts per minute, detecting microscopic surface flaws and dimensional variations that human inspectors would miss. Simultaneously, automated SPC systems plot $\bar{X}$ and R charts in real time, alerting operators the instant a Western Electric out-of-control run pattern appears.

In ergonomics, AI video analytics can track worker postures to calculate RULA and REBA risk scores.

However, an algorithm can only flag an anomaly—it cannot identify the root mechanical cause of tool chatter, nor can it redesign a workstation to fit the physical comfort of human operators. Listening to workers and implementing Kaizen continuous improvements remains human work.

I will now hand the presentation to [Name 5] to review critical risks and ethics."

Slide 9: Critical AI Errors — Hallucinations & Physical Deficits (10:00 – 11:15)

- **On Screen:** Warning graphic illustrating two dangerous failure modes: (1) Technical Hallucination (fabricated material properties and tolerances), and (2) Physical Common Sense Deficit (impossible spatial or temporal assumptions).
- **Speaker 5:**

"Thank you, [Name 4]. While AI tools offer immense power, they exhibit critical vulnerabilities that can produce catastrophic failures if trusted blindly.

The most pervasive issue is 'technical hallucination'. Large language models operate on probabilistic text generation, not empirical validation. This means an AI can state false engineering specifications with supreme confidence.

For example, an engineer querying material limits might receive an AI-generated tensile yield strength or operating temperature that is completely fabricated. If integrated into a manufacturing specification without verification, catastrophic tooling failure or product recall can result.

Furthermore, AI platforms completely lack physical common sense. They do not intuitively grasp gravity, spatial interference, or tool wear dynamics. They may generate an optimized process plan that requires two robotic arms to occupy the exact same physical envelope simultaneously."

Slide 10: Data Privacy, Black-Box Opacity & OIQ Legal Liability (11:15 – 12:30)

- **On Screen:** Legal and ethical matrix: Corporate IP security risks, Neural Network Black-Box Opacity, and the *Code of Ethics of Engineers* under the *Ordre des ingénieurs du Québec* (OIQ).
- **Speaker 5:**

"Beyond technical glitches, AI presents serious legal and ethical challenges.

First is data privacy. Industrial facilities guard proprietary chemical formulas, machining tolerances, and production capacities. Feeding sensitive production parameters into public cloud-hosted AI tools risks leaking trade secrets to competitors.

Second is the 'Black Box' dilemma. Deep neural networks generate recommendations without transparent mathematical proofs. In regulated sectors like aerospace, automotive, and medical manufacturing, regulatory bodies require complete auditability. An engineer cannot justify a safety-critical choice by saying 'the AI recommended it'.

Most importantly, under Quebec law and the OIQ Code of Ethics, software holds zero legal accountability. If an AI-designed assembly process causes worker injury or structural failure, the licensed Professional Engineer (P.Eng.) who stamped the plan carries 100% of the civil and professional liability.

Now, [Name 6] will conclude with the human role and future skills."

Slide 11: Why Human Engineers Are Irreplaceable (Human-in-the-Loop) (12:30 – 13:45)

- **On Screen:** Conceptual flowchart of the Human-in-the-Loop (HITL) Framework: AI Engine (Rapid Permutation, Data Processing, Initial Drafting) → Human Engineer (Verification, Safety Audit, Physical Grounding, Final Stamp).
- **Speaker 6:**

"Thank you, [Name 5]. Everything we have explored points to one clear truth: Generative AI will not replace industrial engineers; it elevates our role through the Human-in-the-Loop paradigm.

In this model, AI serves as an ultra-fast computational co-pilot. It processes millions of data points, generates preliminary schedules, and highlights trends. But the human industrial engineer serves as the indispensable pilot: auditing assumptions, validating physical feasibility, and making final strategic decisions.

Human engineers are irreplaceable when handling unexpected crises. When severe weather shuts down logistics, or a critical machine fails without precedent, AI models have no training data to rely on. Humans excel at rapid adaptation, cross-functional communication, and creative problem-solving.

Furthermore, manufacturing is a human enterprise. Motivating shop-floor teams, fostering safety culture, and building trust cannot be done through an algorithm."

Slide 12: Essential Skills for Engineering Graduates & Conclusion (13:45 – 15:00)

- **On Screen:** Graduate Competency Blueprint:

1. Practical Programming & Data Literacy (Python, SQL)
2. Core Operations Research Math (Optimization, SPC, Queuing)
3. AI Prompting & Output Verification Auditing
4. Interpersonal Communication, Leadership & Professional Ethics

- **Speaker 6:**

"To succeed in this evolving landscape, undergraduate engineering students must develop a balanced set of competencies.

First, practical data fluency: learning Python and data handling alongside traditional tools like Excel.

Second, mastering foundational engineering mathematics. The stronger your understanding of operations research, probability, and statistical quality control, the faster you will detect AI hallucinations and mathematical errors.

Third, honing communication and leadership abilities. Bridging the gap between executive strategy, data science, and shop-floor technicians is something technology will never replicate.

In conclusion, AI is a powerful multiplier for industrial engineering, but human judgment, ethical accountability, and physical common sense remain the bedrock of production systems.

Thank you very much for your attention. Our team is now ready to take any questions."